

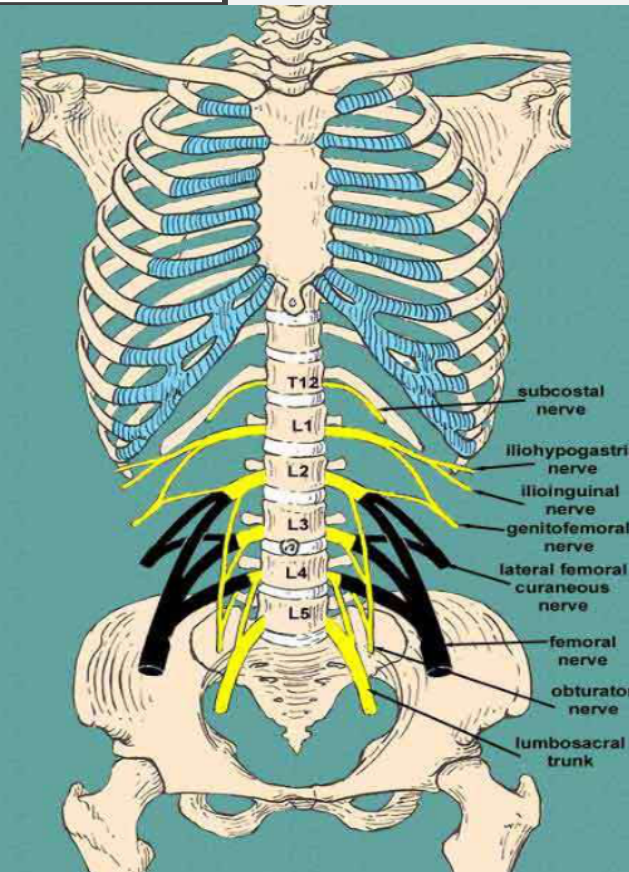
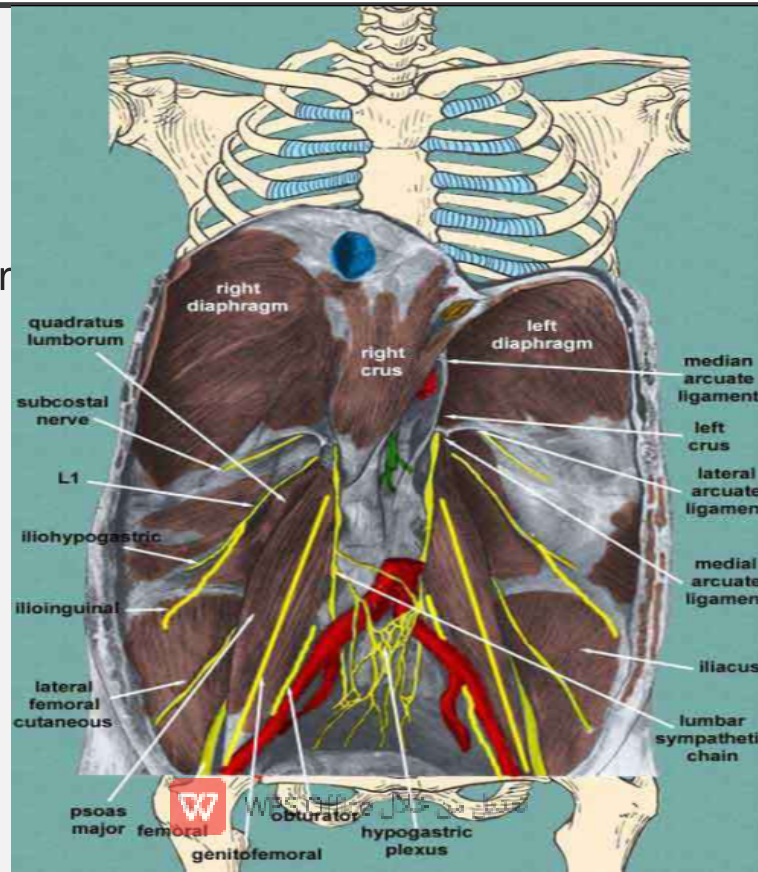
POSTERIOR ABDOMINAL WALL

BY: M. G. HAMMAD



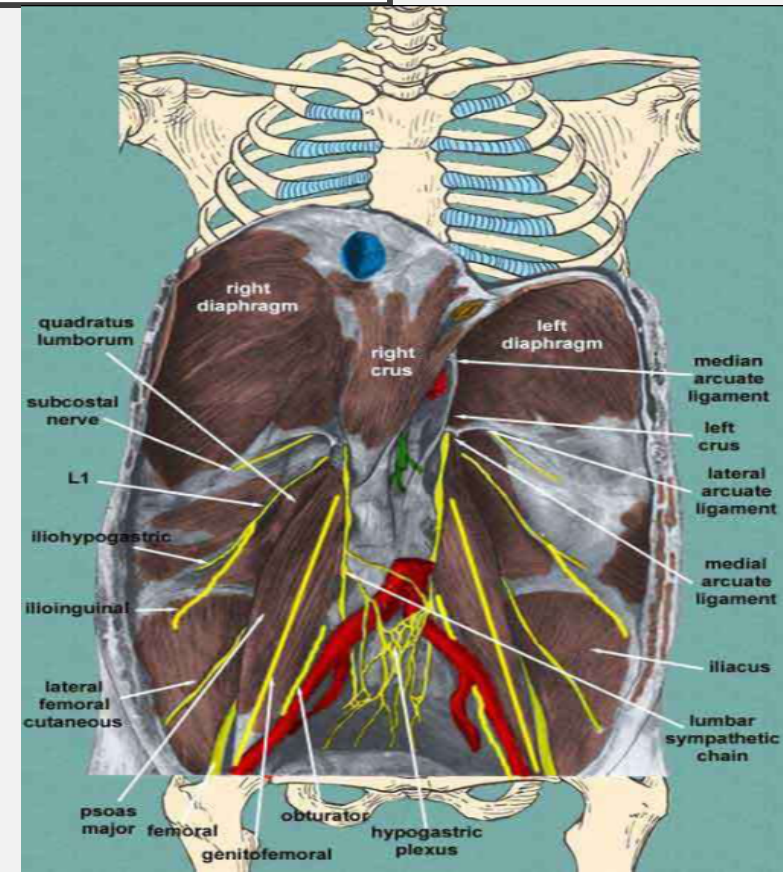
POSTERIOR ABDOMINAL WALL

- PAW supported by:
 - ✓ Twelfth rib.
 - ✓ Twelfth thoracic vertebra.
 - ✓ Lumbar vertebrae.
- It contains:
 - ❖ Muscles.
 - ❖ Nerve plexus.
 - ❖ Blood vessels.
 - ❖ Lymphatics.



POSTERIOR ABDOMINAL WALL

- Extends from the 12th rib above to the pelvic brim below.
- Supported by bones, muscles, and fasciae.
- It supports retroperitoneal organs, vessels, and nerves.
- The posterior abdominal wall is constructed as:
 - i. Bony.
 - ii. Muscular part.
 - iii. Fasciae. (Psoas fascia, Fascia iliaca, & Thoraco-lumbar fascia).



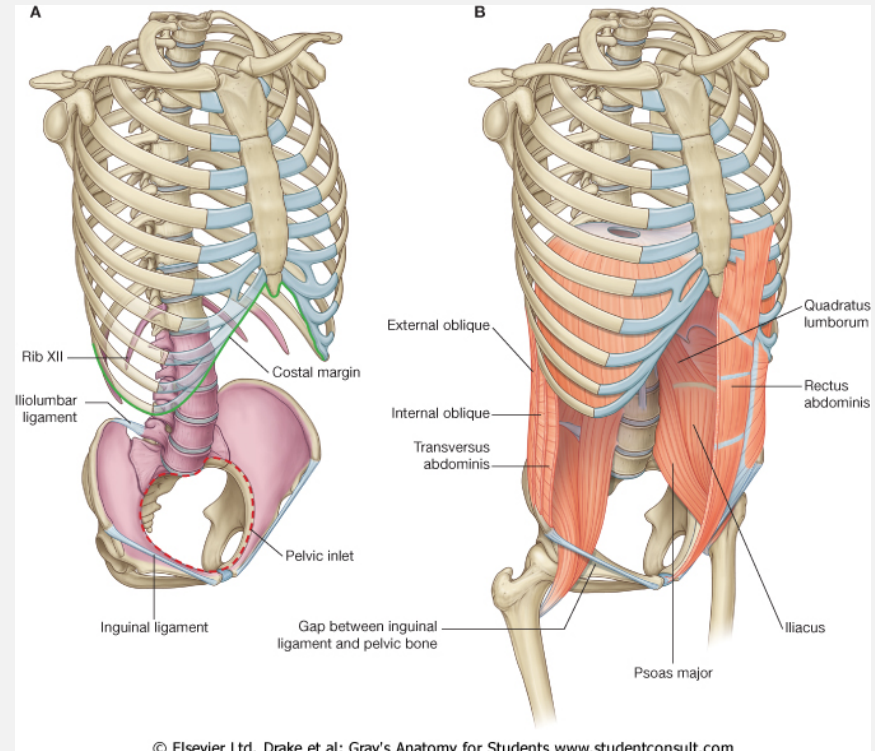
MUSCLES OF THE POSTERIOR ABDOMINAL WALL

- 4 muscles:
- ✓ Diaphragm.
- ✓ psoas muscles.
- ✓ Iliacus.
- ✓ Quadratus lumborum.



PARAVERTEBRAL GUTTERS

- Deep grooves on both sides of VC.
- It floored by:
 - ❖ Above:
 - Diaphragm.
 - Psoas muscles.
 - Quadratus lumborum.
 - ❖ Below:
 - Iliacus muscle.



MUSCLES OF PAW:

PSOAS MAJOR

❑ Origin:

- Transverse processes , sides of bodies & intervertebral discs of 12th thoracic & the 5 lumbar vertebrae.

❑ Insertion:

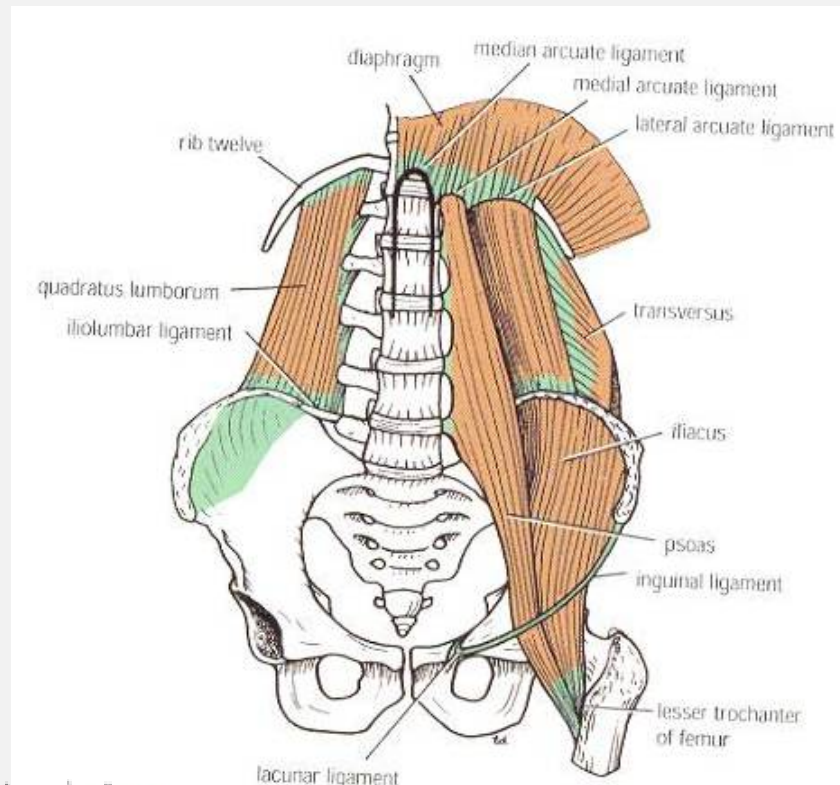
- With iliacus into the lesser trochanter of femur.

❑ Nerve supply:

- Lumbar plexus(L1,2,3).

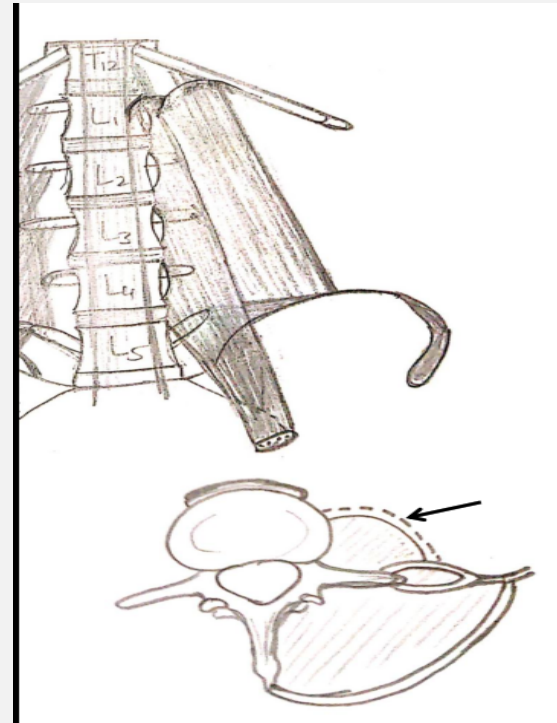
❑ Action:

- Flexes thigh on trunk at hip joint & if thigh is fixed , it flexes the trunk on thigh , as in sitting up from lying position.



PSOAS FASCIA

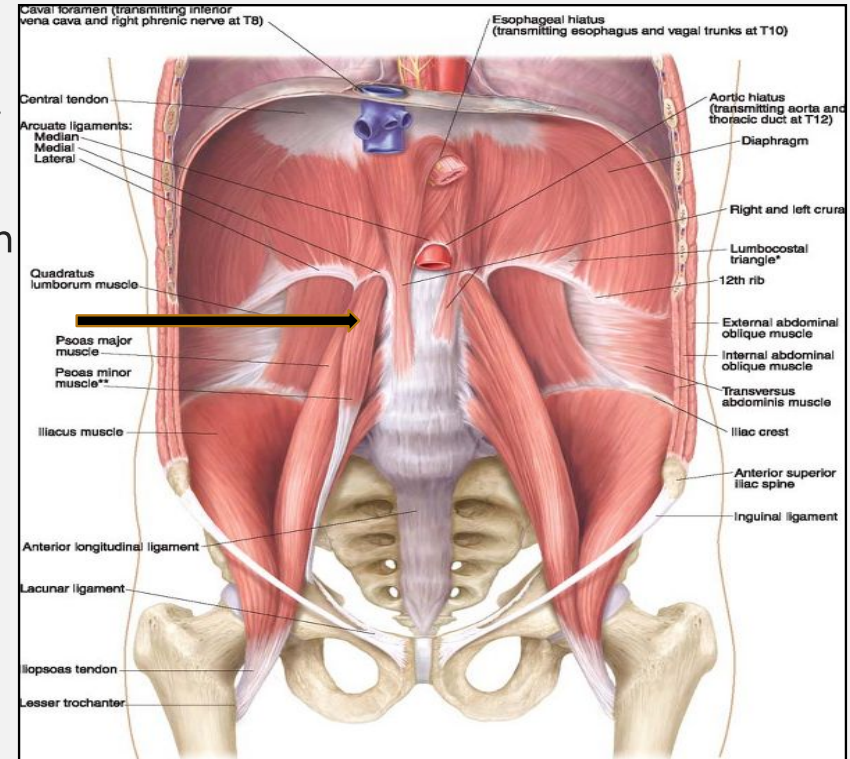
- Covers psoas muscle.
- Attachment:
 - ❑ Superior:
 - Body & TP of L1 (medial arcuate ligament).
 - ❑ Lateral:
 - Fused with anterior layer of thoracolumbar fascia.
 - ❑ Medial:
 - Bodies of lumbar vertebrae & intervertebral discs.
 - ❑ Inferior:
 - Fused with fascia covering iliacus.
- ❖ The fascia forms sheath around psoas muscle called psoas sheath.



MUSCLES OF PAW:

PSOAS MINOR

- ❑ It is a small unimportant muscle, absent in 40% of people, if it is present it lies in front of psoas major.
- ❑ Origin:
 - Sides of bodies of T12 & L1 vertebrae & disc between them.
- ❑ Insertion:
 - Iliopectineal eminence & pectineal line.
- ❑ Nerve supply:
 - Ventral ramus of L1 N.
- ❑ Action:
 - Flexion of lumbar part of VC.



MUSCLES OF PAW:

ILIACUS

❑ Origin:

- Iliac fossa.

❑ Insertion:

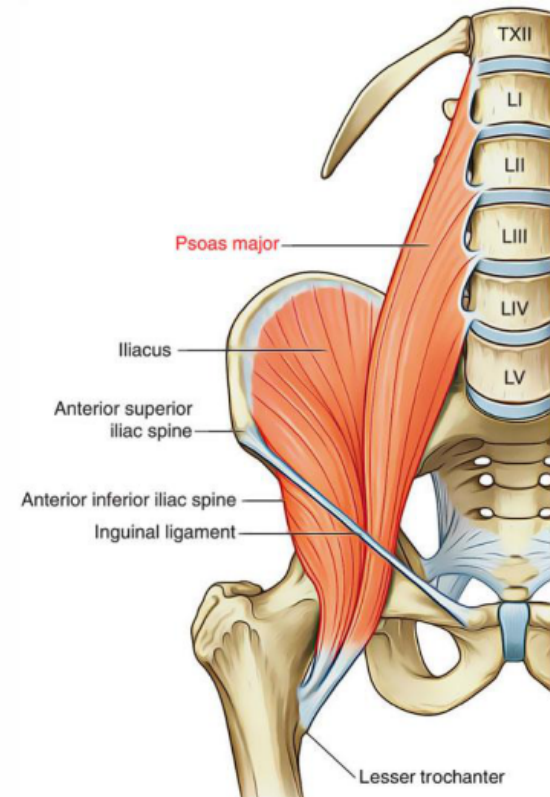
- ❖ With the psoas major tendon, into the lesser trochanter of femur.
- ❖ The combined muscles are called iliopsoas.

❑ Nerve supply:

- Femoral nerve.

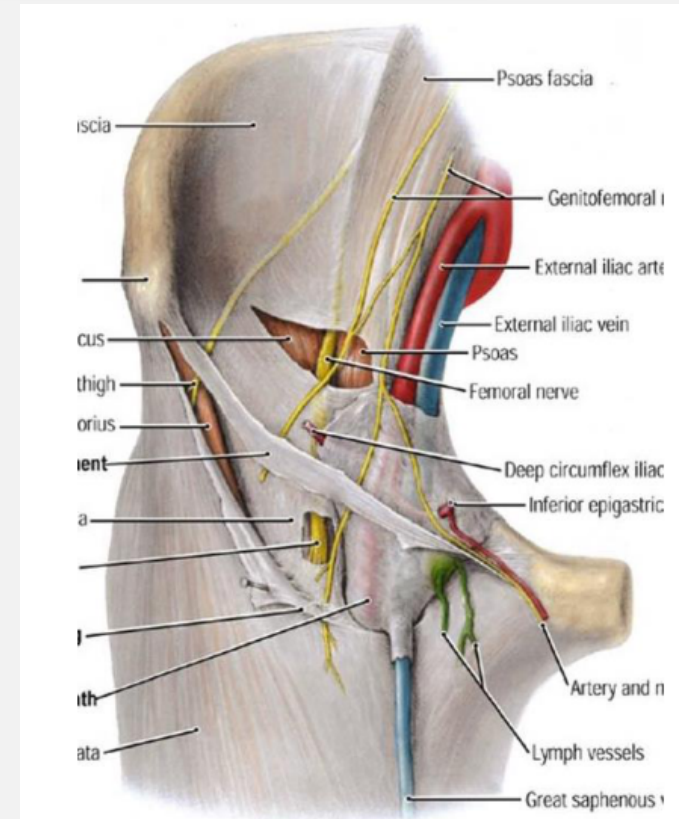
❑ Action:

- Flexes thigh on trunk at hip joint & if thigh is fixed , it flexes the trunk on thigh, as in sitting up from lying position.



ILIAC FASCIA

- Covers the iliacus muscle.
- Attachment:
- ❑ Medial:
 - Merges with psoas fascia & attached to iliopectineal eminence.
- ❑ Lateral:
 - Margins of iliac crest
- ❑ Inferior:
 - Passes deep to inguinal ligament & forms posterior layer of femoral sheath.



MUSCLES OF PAW: QUADRATUS LUMBORUM

❑ Origin:

- ❖ Iliolumbar ligament , iliac crest , tips of transverse processes of lower 2-3 lumbar vertebrae.

❑ Insertion:

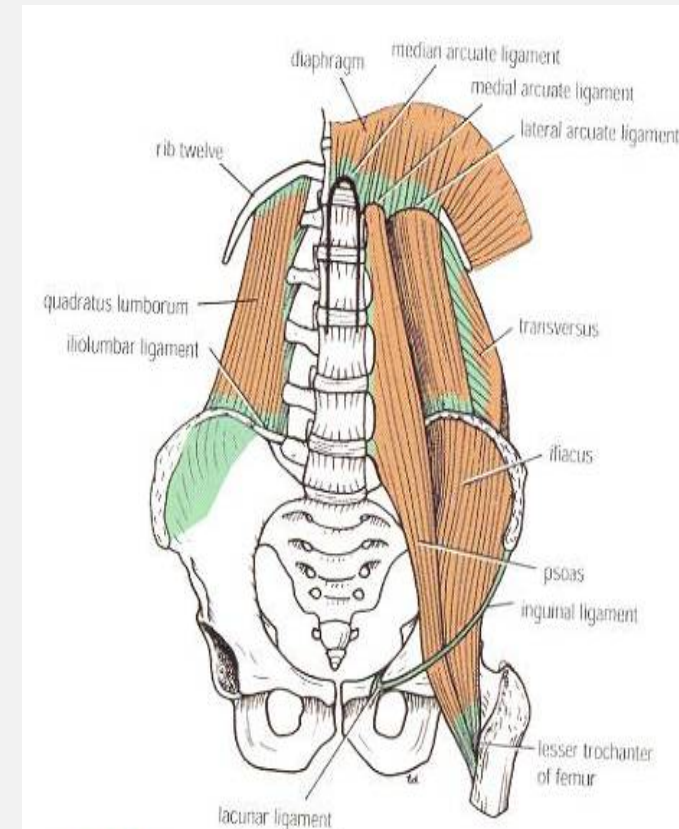
- ❖ lower border of 12th rib & upper 4 lumbar vertebrae.

❑ Nerve supply:

- Lumbar plexus(L1,2,3,4).

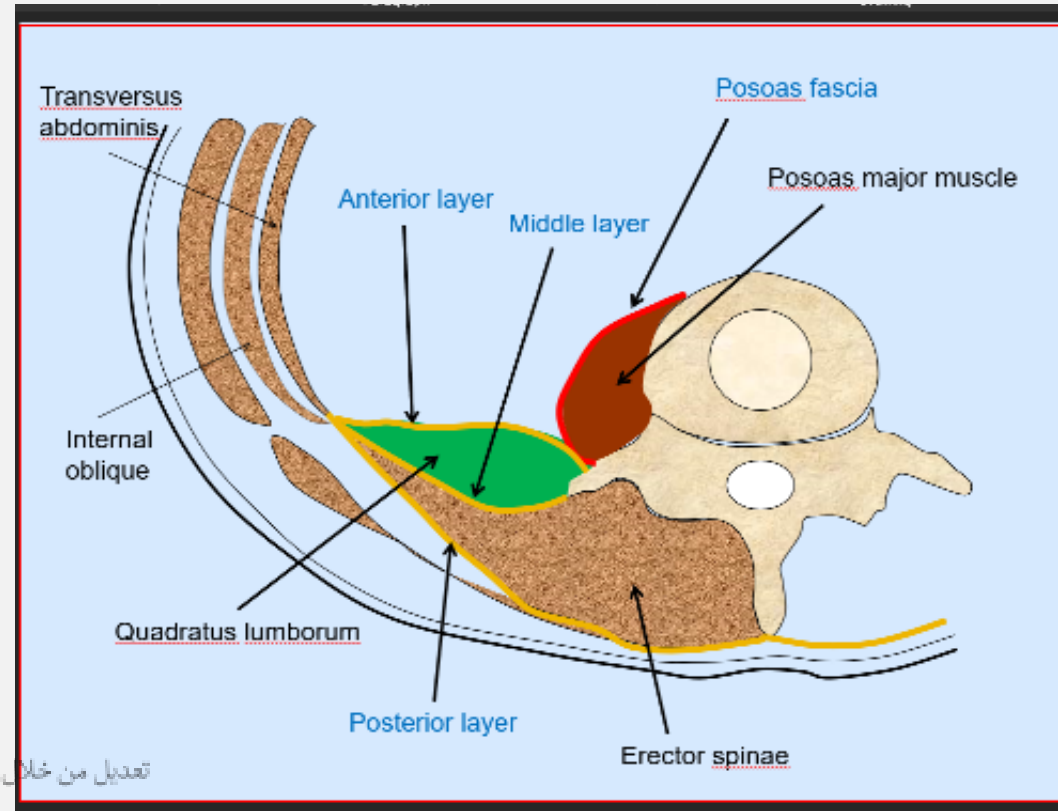
❑ Action:

- ❖ Fixation of 12th rib to help contraction of diaphragm during deep inspiration, depresses 12th rib during forced expiration, lateral flexion of vertebral column on the same side.
- ✓ It is covered by lumbar fascia , which is thickened above forming lateral arcuate ligament, and below forming iliolumbar ligament.



THORACOLUMBAR FASCIA

- ❑ Extends from the back of sacrum to the neck.
- ❑ Passes binds to erector spinae muscle.
- ❑ It attaches to vertebral spines.
- ❑ Encloses the quadratus lumborum.
- ❑ Very strong in the lumbar region.
- ❑ Laterally it gives origin to internal oblique and transversus abdominis muscles.
- ❑ Its lumbar part medially splits into three layers.

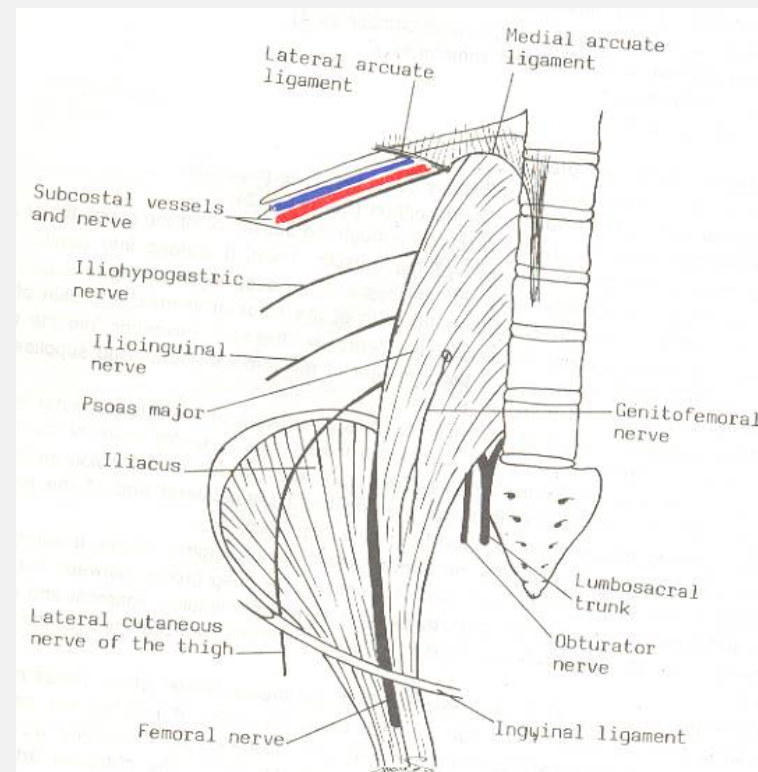


PAW



RELATION OF PSOAS MAJOR

- Inside its substances:
 - ✓ Lumbar plexus is formed.(L1,2,3,4).
- Anteriorly:
 - ✓ It is covered by psaos fascia, psaos minor and is related to: kidney, ureter, renal vs., gonadal vs., genitofemoral N.
- Posteriorly :
 - ✓ Lumbar transverse processes + medial edge of Q.L.
- Medially :
 - ✓ Lumbar vertebral bodies, lumbar vessels & sympathetic trunk.
- Laterally :
 - ✓ These Ns. emerge from its lateral border : iliohypogastric N., ilioinguinal N., lateral cutaneous N of thigh & femoral N., which lies in deep groove between psaos major & iliacus.



THAT IS ALL

